

Nigel Loh

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EDUCATION

University of Toronto Toronto, Canada
Bachelor of Science in Computer Science and Math (Co-op Program) (GPA: 3.96 / 4.0) Expected June 2027

- Coursework: Operating Systems, Machine Learning, Deep Learning, Data Structures, Algorithms, Databases, Computer Organization

WORK EXPERIENCE

Capital One | *NestJS, TypeScript, Angular, AWS, Jenkins, PostgreSQL, Claude Code* McLean, Virginia
Software Engineer Intern June to August 2026

- Architected a self-service questionnaire platform letting **15+** internal banking teams configure and deploy dynamic intake forms through an Angular interface without engineering support, cutting workflow launch time **95%** (weeks to hours)
- Built REST APIs in NestJS and TypeScript over PostgreSQL for questionnaire schemas and response persistence, modeling reusable sections as UUID references so teams composed forms without duplicating content, sustaining **3M+ API calls** per day
- Designed a **tagging system** replacing the multi-table scan behind combined department and creator filters, resolving questionnaire lookups against a single table; shipped through Jenkins CI/CD to AWS ECS with audit logging

PointClickCare | *Python, MySQL, Azure, OpenAI, LangChain* Toronto, Ontario
Machine Learning/Data Engineer Intern May to August 2025

- Built a RAG pipeline over **10M+** clinical notes and **5M+** assessment questions using OpenAI text-embedding-3-small, Azure AI Search, and LangChain, cutting a chart review the team reported taking weeks down to minutes
- Validated retrieval quality by manually reviewing a random sample of **3,000+** question matches, tuning the similarity threshold where matches returned too loose
- Built high throughput pipelines with Pandas, NumPy, and MySQL to ingest and preprocess **20M+** clinical records

PROJECTS

Settlers of Solana (Hack the North Winner) | *Rust, Solana, TypeScript, React, Node.js, OpenAI API*

- Named a **Finalist** and won **Best Use of Solana (\$7,000 in prizes)** at **Hack the North (Canada's largest hackathon, 1,000+ participants)** for an **adaptive free-market economy of 100+ LLM agents** with no fixed jobs, wages or prices, each agent choosing a trade every round, posting limit orders and borrowing against its goods, so prices emerge from supply and demand alone
- Built the settlement layer as a **Rust** program on **Solana** holding every balance, the order books and the bank's balance sheet, clearing a **100-order auction atomically in 12,717 compute units (under 1% of budget)** by verifying an off-chain sort in one pass, with coins minted only by lending

Dance Dance Revolution Keyboard (UofTHacks Winner) | *C++, Python, Arduino*

- Led a team of four to build a working dance pad keyboard, writing C++ Arduino firmware that streamed key down and key up events over USB to a Python layer decoding steps as **Braille** chords, with a coalescing window so presses landing milliseconds apart resolved to one character instead of several
- Won **Best Hardware** at UofTHacks13 among 500+ participants by creating an interactive input device that combines movement with productivity

MammoNet | *Python, PyTorch, Django, Next.js*

- Led a team at a **bioinformatics hackathon** to build a breast cancer classification system with a convolutional neural network, reaching **96% test accuracy** on microscopy images
- Served the model behind a **Django REST API** for uploads, preprocessing, and real time inference, wired to a **Next.js** interface with **heatmap visualizations** of malignancy indicators

TECHNICAL SKILLS

- Languages:** Python, TypeScript/Javascript, Java, C/C++, Rust, Assembly, SQL
- Frameworks & Databases:** Node.js, NestJS, React, Angular, FastAPI, PyTorch, PostgreSQL, MySQL, Redis
- Cloud & Tools:** AWS (ECS, EC2), Azure, Docker, Jenkins CI/CD, Git, Linux, Jest, Cucumber, Claude Code